

The LU Decomposition of a Matrix Examples 1

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Recall from [The LU Decomposition of a Matrix](#) page that if we have an $n \times n$ matrix A , then provided that under Gaussian Elimination, an upper triangular matrix U can be produced without pivoting, then there exists another matrix L that is lower triangular such that $A = LU$.

We will now look at some concrete examples of finding an LU decomposition of a matrix.

Example 1

Find an LU decomposition for the matrix $A = \begin{bmatrix} 3 & 1 \\ 4 & 2 \end{bmatrix}$.

We will start by applying Gaussian Elimination to get a row equivalent form of A that is upper triangular. We do this by the elementary row operation $R_2 - \frac{4}{3}R_1 \rightarrow R_2$ to immediately obtain an upper triangular matrix, U :

$$U = \begin{bmatrix} 3 & 1 \\ 0 & \frac{2}{3} \end{bmatrix}$$

(1)

Now our corresponding lower triangular matrix L is going to have 1's along its main diagonal.

The entry below the main diagonal is obtained as the inverse row operations applied to . In this case, we have to obtain:

(2)

Therefore our decomposition of is:

(3)

Note that we will only be using the elementary row operations of addition/subtraction of a multiple of one row to another, and so the inverse operations will always be the negative of the multipliers used in performing Gaussian Elimination to get to .

(4)

Example 2

Find an decomposition for the matrix .

Once again, we begin by using Gaussian Elimination. We take to get:

We now take to get:

(5)

Lastly we take to obtain our upper triangular matrix :

(6)

Our corresponding lower triangular matrix will once again have 's along the main diagonal, and the entries underneath the main diagonal are obtained from the corresponding inverse operations. Thus:

(7)

Therefore an decomposition for is:

(8)

Note in this particular example that the third row of is all zeroes. This implies that itself is noninvertible.

(9)